

Question:

Six $36\,\Omega$ resistors are connected in a combination of series and parallel between points X and Y (as shown in the question diagram). What is the total resistance between X and Y?

Answer:

C. $12\,\Omega$

Explanation:

The diagram shows a three-branch layout between points **X** and **Y**:

• **Top branch:** Two $36\,\Omega$ resistors in **series** → total = **$72\,\Omega$**

• **Middle branch:** Two $36\,\Omega$ resistors in **parallel** →

$$\frac{1}{R} = \frac{1}{36} + \frac{1}{36} = \frac{2}{36} = \frac{1}{18} \Rightarrow R = 18$$

• **Bottom branch:** Two $36\,\Omega$ resistors in **series** → total = **$72\,\Omega$**

Now, the three branches ($72\,\Omega$, $18\,\Omega$, $72\,\Omega$) are connected in **parallel**:

$$\frac{1}{R_{\text{total}}} = \frac{1}{72} + \frac{1}{18} + \frac{1}{72} = \frac{1 + 4 + 1}{72} = \frac{6}{72} = \frac{1}{12} \Rightarrow R_{\text{total}} = 12$$

Guidance for Students:

- Always simplify **series resistors first** — add their resistances directly.
- Then combine **parallel resistors** using the reciprocal formula.
- For more complex arrangements, **redraw the circuit** to isolate each path from point X to point Y.
- Don't forget that two equal resistors in parallel halve the value:

$$R_{\text{parallel}} = \frac{R}{2}$$

Revision Tips:

- • Two resistors in series: **add the resistances**.
- • Two resistors in parallel:
$$R = \frac{R_1 R_2}{R_1 + R_2}$$
- • Identify parallel branches and simplify step-by-step.
- • Use the reciprocal rule when more than two resistors are in parallel.
- • Draw or trace each branch clearly — visuals really help.